

Horizon-Dependent Value of Geodetic Context in Operational Earthquake Forecasting: A Leakage-Free, Multi-Region Study with Pre-Registered Negative Results

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Abstract—Operational earthquake forecasting (OEF) issues calibrated conditional probabilities of future seismicity; the Epidemic-Type Aftershock Sequence (ETAS) model is its de-facto benchmark, and no machine-learning temporal point process has been shown to beat a well-fit ETAS prospectively under fair testing. We build a leakage-free, multi-region OEF testbed on a global moment-magnitude catalog ($M \geq 5$, 1990–2026), score every model with the community-standard CSEP framework (pyCSEP) in information gain per earthquake (IGPE, nats), and pre-register ship/no-ship rules with shuffled-label negative controls. Within it we report four findings, each artifact-backed and honestly framed. *First*, the value of a GNSS-strain geodetic context covariate, added to a Hawkes-structured neural point process, is *horizon-dependent*: at the 1–7 day operational horizon the strain-conditioned model does not beat tiled ETAS (mean IGPE -0.053 over eight weekly windows, only 4/8 positive), whereas at a 30 day horizon it beats ETAS robustly (global mean IGPE $+0.115$ over 2166 events, 9/10 windows positive, paired- t lower bound $+0.087$; positive in all four high-seismicity regions). The mechanism is explicit: the calibrated model deploys a time-flat geodetic *background*, not a triggering model, so it wins where the background dominates the integral (30 days) and loses where ETAS triggering dominates (1–7 days). *Second*, a convex log-score-optimal stack of ETAS-family variants earns a small but significant global gain ($+0.011$ over 751 events) that does *not* generalize to the canonical active margins; under the pre-registered rule it is a no-ship, and a temporally adaptive variant is refuted as a multiple-comparison artifact. *Third*, ETAS adds skill over a stationary smoothed-seismicity Poisson null only where there is triggering to exploit, quantifying the high-versus-low-seismicity bias rather than asserting it. *Fourth*, the binding 1–7 day consistency failure is count over-dispersion, not spatial shape: a catalog-based (branching-simulation) number test passes a window the Poisson number test rejects, and the frozen-intensity forecast systematically under-counts by $\approx 28\%$ because it omits within-window secondary triggering. We conclude that base tiled ETAS is at or near the practical ceiling for mean-rate 1–7 day IGPE over global $M \geq 5$, and that the honest place for a geodetic covariate is a longer, background-dominated outlook.

Index Terms—operational earthquake forecasting, ETAS, epidemic-type aftershock sequence, neural temporal point process, geodetic strain, GNSS, CSEP, pyCSEP, information gain, prospective evaluation, over-dispersion, reproducibility.

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Preprint, 2026. Code, configurations, provenance manifests, and the committed daily/back-analysis artifacts are released under MIT at https://github.com/fsantibanezleal/CAOS_SEISMIC; a static forecast viewer is live at <https://seismic.fasl-work.com>. This is an independent research and education tool; it is not an alarm system and must not be used for life-safety decisions.

I. INTRODUCTION

EARTHQUAKES cannot be predicted deterministically: whether a small rupture cascades into a great earthquake depends on unmeasurably fine details of the crust [1]. What *can* be done, and is done operationally, is *forecasting*: issuing a probability strictly in $(0, 1)$ for a region, a magnitude band, and a short horizon. Following the ICEF definition [2], a prediction is a deterministic statement that an event will or will not occur, whereas a forecast reports a probability with uncertainty. Short-term probabilities of a large event may vary over orders of magnitude relative to background yet typically remain small in absolute terms (well under a few percent per day) [2]. This distinction is not pedantic: the documented harm in the field is *false reassurance*, and every number here is a conditional probability shown next to its long-term baseline, scored prospectively, and never rendered as an alarm.

The de-facto short-term baseline is the Epidemic-Type Aftershock Sequence (ETAS) model [3], [4], a self-exciting Hawkes point process whose conditional intensity superposes a stationary background on the summed, Omori–Utsu-decaying, Gutenberg–Richter-weighted offspring of every past event. Operational systems such as the USGS aftershock forecast [5], [6] and OEF-Italy [7] run ETAS-class models as scheduled services and publish calibrated probabilities with uncertainty. A decade of prospective next-day forecasting over California confirms that no single model dominates: the short-term-clustering models excel during aftershock sequences, and ETAS is the consistent generalist [8].

Against this baseline, machine learning has repeatedly promised more and, under fair testing, repeatedly failed to deliver a prospective gain. The EarthquakeNPP benchmark [9] evaluated five modern neural spatio-temporal point processes on California with strict chronological splits and CSEP consistency tests, and found that *none* outperformed ETAS, precisely on the spatial test where forecasting value lives; the apparent advantages of earlier neural work trace to a data-leakage flaw (non-chronological splits) and to excluding the large sequences. The canonical cautionary tale is a deep network for aftershock patterns [10] whose reported skill was matched by a two-parameter logistic regression on a single feature [11], with the area under the ROC curve exposed as blind to the calibration a forecast must publish. Where learned models *can* help ETAS, the gains come from two real gaps, not

from network depth: ingesting covariates ETAS cannot easily absorb, and learning spatial anisotropy [12], [13].

This paper studies one such covariate, *geodetic strain* from GNSS, and asks not whether it helps but *at which horizon*. Geodesy is an established input to long-term global rate models: the GEAR1 model blends smoothed seismicity with a strain-rate model [14], and prospective global hybrids find geodetic strain adds skill globally while its regional contribution is weaker [15]. We port that intuition into a short-term OEF testbed and measure it prospectively. Our contributions are:

- 1) A leakage-free, multi-region OEF testbed with a strict forecast clock, CSEP/pyCSEP scoring in IGPE, pre-registered ship rules, and shuffled-label negative controls (Sec. III).
- 2) The *horizon-dependent* value of a GNSS-strain context covariate: no gain at 1–7 days, a robust and regionally consistent gain at 30 days, with a mechanistic explanation (Sec. IV-D).
- 3) Pre-registered *negative* results: ETAS-family stacking that does not generalize across regions, and the 1–7 day mean-rate IGPE ceiling (Sec. IV-B).
- 4) A methodological result: over-dispersion-honest, catalog-based consistency scoring, and the $\approx 28\%$ secondary-cascade under-count of a frozen-intensity forecast (Sec. IV-C).

Every result is backed by a committed, reproducible artifact; negative and null results are reported, not hidden.

II. DATA

The catalog spine is USGS ComCat (FDSN event service, public domain), homogenized to moment magnitude M_w . A per-network station-pair regression to M_w is not always available, so the Scordilis [16] global $m_b \rightarrow M_w$ and $M_s \rightarrow M_w$ orthogonal-regression relations are used as a documented fallback, with per-event provenance recorded; this recovers a $6\times$ larger conditioning catalog (from 1.65×10^4 to 9.7×10^4 events, $M \geq 4.5$, 1990–2026) at the cost of a bounded, audited b -value inflation that the per-tile fit largely removes (naive global $b \approx 1.59$; per-tile production $b \approx 1.34$). The magnitude of completeness M_c and the Gutenberg–Richter b -value are estimated on rolling windows [17], [18]. Declustering, where needed for the stationary background, uses stochastic declustering [19] and the nearest-neighbour method [20]. The geodetic covariate is the GNSS strain rate from the Nevada Geodetic Laboratory MIDAS velocity field [21] (20,168 stations after an endpoint and column-parse fix), gridded to a coarse global field; strain is high on active margins (Japan 95, California 154, Chile 64 nstrain/yr) and low in stable interiors (Central US 3.6, Western Europe 5.6 nstrain/yr). Plate-boundary geometry is PB2002 [22]. Raw catalogs, features, and model weights are rebuildable from the code and provenance manifests and are not versioned; the compact daily and back-analysis artifacts are.

III. METHODS

A. Baseline models

ETAS. The space–time conditional intensity [4] is

$$\lambda(t, x, y | \mathcal{H}_t) = \mu(x, y) + \sum_{i: t_i < t} K e^{\alpha(M_i - M_0)} \left(1 + \frac{t - t_i}{c}\right)^{-p} f(x - x_i, y - y_i | \mathcal{H}_t) \quad (1)$$

with the Ogata inverse-power spatial kernel $f(x, y | M_i) = \frac{q-1}{\pi\zeta^2} (1 + r^2/\zeta^2)^{-q}$, $\zeta = D e^{\gamma(M_i - M_0)}$. Parameters are fit by maximizing the point-process log-likelihood

$$\ln L = \sum_i \ln \lambda(t_i, x_i, y_i | \mathcal{H}_{t_i}) - \int_0^T \int_A \lambda(t, x, y | \mathcal{H}_t) dx dy dt, \quad (2)$$

subject to two stability gates: finite branching ($\alpha < \beta$, $\beta = b \ln 10$) and subcriticality (branching ratio $n < 1$). Because a single global ETAS over a 10^5 -event catalog is both intractable and physically wrong (subduction and stable interior obey different kinematics), we fit a *regime-tiled* ETAS: 195 ETAS tiles are fit and 265 sparse tiles fall back to their smoothed null, on $M_c = 5.35$, $b = 1.337$.

Smoothed-seismicity Poisson null. The stationary background $\mu(x, y) = \sum_i K_{d_i}(r)$ uses the Helmstetter–Kagan–Jackson adaptive power-law kernel with bandwidth set by the distance to the n -th nearest neighbour [23]. It serves both as the ETAS background field and, crucially, as the *mandatory stationary null* any time-dependent model must beat. **Reasenbergs–Jones** [5], [6] is a transparent aftershock-only fallback and sanity check.

B. Neural challenger with geodetic context

The challenger is a context-conditioned temporal point process (`context_tpp`): a CNN encodes a gridded covariate field (here the `gnss_strain_rate` channel; other geophysical channels are honestly zero-filled) into a Hawkes-structured conditional intensity, in the spirit of RECAST [12] and FERN [13] (keep the additive background plus summed-triggering skeleton, replace fixed kernels with learned maps). It is trained on the global catalog on a single laptop GPU. A fit-time scalar `rate_cal` anchors the integrated one-day forecast rate to the training rate on the same multi-resolution grid the evaluation uses; without it the background head overpredicts the absolute rate by $\approx 490\times$ (a failure we diagnose and report, Sec. IV-D). The challenger is a *gated* model: it reaches the public map only if it beats ETAS prospectively *and* is calibrated; until then, ETAS ships.

C. Evaluation: CSEP, IGPE, and the forecast clock

Skill is established with the community CSEP framework via pyCSEP [24]. Consistency (calibration) is tested with the number (N), magnitude (M), spatial (S), and conditional-likelihood (CL) tests [26], [27], built on the Poisson joint log-likelihood over space–magnitude bins. *Passing consistency is necessary but not sufficient*: skill requires winning a compari-

TABLE I
ETAS IGPE VS. THE POISSON NULL, 60-DAY LEAKAGE-FREE
BACK-ANALYSIS. SKILL CONCENTRATES IN THE HIGH-SEISMICITY,
TRIGGERING-RICH REGIME.

View	Class	IGPE (1 d)	IGPE (7 d)
Japan	high	+0.072	+0.062
Chile	high	+0.0015	+0.0024
California	high	~ 0	+0.0016
New Zealand	high	~ 0	-0.001
Central US	low	0.0	0.0
Western Europe	low	0.0	0.0
Eastern Australia	low	0.0	0.0

son test against both the Poisson null and ETAS, measured as information gain per earthquake (IGPE, nats):

$$I_N(A, B) = \frac{1}{N} \sum_{i=1}^N (\ln \lambda_A(k_i) - \ln \lambda_B(k_i)) - \frac{\hat{N}_A - \hat{N}_B}{N}, \quad (3)$$

tested with the paired t -test $T = I_N/(s/\sqrt{N}) \sim t_{N-1}$ and corroborated by the Wilcoxon signed-rank W -test [28]. ROC/AUC is deliberately *banned* as a skill metric: it is invariant to calibration and, on rare per-cell-per-day tasks, degenerates into a region classifier (the DeVries trap) [10], [11]. Proper scoring rules [32] and a Molchan/area-skill view [33] are reported only as communication aids.

Leakage is structural. A strict forecast clock hands the model only the catalog slice up to issue time; each forecast is logged immutably with its exact input state, so any past forecast is byte-reproducible and scored against the catalog as it was at issue time. The authoritative measure is a *pseudo-prospective back-analysis*: fit the model family at the earliest cutoff, then *recondition* forward across weekly, non-overlapping, fully-scorable windows (refresh the parents, hold the parameters and background), scoring IGPE per window and per regional view. Reconditioning reproduces full-refit-every-day to 0.05% median deviation while running $13\times$ faster. Ship/no-ship rules are *pre-registered* before each run, and a shuffled-label negative control guards against over-fitting.

IV. RESULTS

A. ETAS adds skill only where there is triggering to exploit

The first adversarial question is whether ETAS only looks good because it over-fits high-seismicity margins. A 60-day, leakage-free, per-country back-analysis (1-day horizon, IGPE vs. the Poisson null) answers it quantitatively (Table I): ETAS adds skill in the densest triggering regime (Japan, +0.072) and collapses to the null in low-seismicity interiors (Central US, Western Europe, Eastern Australia, all 0.0). A single-window benchmark tells the same story at larger absolute scale (global +3.71 nats vs. the null), and shows that Reasenber–Jones alone is no better than the null. Low-seismicity 0.0 is not a failure; it is the correct statement that a self-exciting model has nothing to add without aftershock sequences.

TABLE II
PRE-REGISTERED MULTI-REGION GATE FOR THE ETAS-FAMILY STACK,
7-DAY HORIZON, 14 WEEKLY WINDOWS. GLOBAL GAIN IS REAL;
REGIONAL GENERALIZATION FAILS THE RULE.

View	mean IGPE vs. base	windows +	paired- t CI
Global	+0.0111	13/14	excludes zero (+)
Chile	-0.0001	3/10	crosses zero
Japan	-0.0008	8/13	crosses zero
California	+0.0077	3/3 (noise)	crosses zero
New Zealand	-0.0008	0/3	crosses zero

B. ETAS-family stacking: a global gain that does not generalize

The only prospective lever the literature supports over a single well-fit ETAS is a score-weighted ETAS-family ensemble, and only by a small, often non-significant margin [30], [31]. A naive equal-weight pool of heterogeneous models under-performs ETAS because averaging in the null dilutes the triggering signal; so we build a convex, log-score-optimal stack of three *ETAS-family* variants (base tiled, short-memory, long-memory), with the smoothed null entering only as each member’s background, weights shrunk toward the base vertex, and a pre-registered rule requiring a positive, CI-excludes-zero gain on the global view *and* in at least two high-seismicity regions at the 7-day horizon.

The 14-window, 7-day back-analysis (Table II) earns a real, significant *global* gain (+0.011 over 751 events, 13/14 windows positive, paired- t CI excludes zero), carried by the long-memory variant capturing late-aftershock structure the base cutoff misses. But only 1/4 high-seismicity regions is positive (California, on 3 events, noise). Under the pre-registered rule this is a **no-ship**. A temporally adaptive variant appears to rescue it (3/4 regions positive) but is refuted on scrutiny: it *reduces* the global gain, and the regional “improvements” are sign-flipping noise on 3–54 events (no regional CI excludes zero), a textbook multiple-comparison artifact. The honest reading: the regional 7-day $M \geq 5$ gate is under-powered; the global view is the only adequately powered test, and there the static stack already gives the answer.

C. The 1–7 day failure is over-dispersion, not shape

Why does the base ETAS N -test fail on active windows? Regional seismicity is over-dispersed relative to Poisson (variance $\gg$ mean, from clustering), so the Poisson number test over-rejects vigorous-but-plausible sequence counts. We add a catalog-based number test [25], [29]: an ETAS branching-process forward simulation produces the honest in-window $M \geq M_c$ count distribution. On a 30-day global window (Table III), the frozen-intensity forecast predicts $N_{\text{fore}} = 64.3$ (background 60.1 + first-generation triggering 4.2), the full branching cascade lifts the mean to 82.1 with Fano factor 1.79, and the observed 96 is consistent: the Poisson N -test fails at quantile 0.0001, while the catalog-based test passes at 0.133. Two honest findings follow. The apparent under-forecast is over-dispersion plus secondary cascade, not a rate bias. And the frozen-intensity forecast *systematically under-counts* by $\approx 28\%$ because it omits within-window secondary triggering;

TABLE III
OVER-DISPERSION-HONEST CONSISTENCY, 30-DAY GLOBAL WINDOW.
THE CATALOG-BASED TEST PASSES WHERE THE POISSON TEST
OVER-REJECTS.

Quantity	Value
N_{fore} (frozen intensity)	64.3
simulated mean (full cascade)	82.1 (+27.6%)
Fano factor (Var/mean)	1.79
observed $M \geq M_c$	96
Poisson N-test	$q = 0.0001$ (fail)
catalog-based N-test	$q = 0.133$ (pass)

TABLE IV
STRAIN-CONDITIONED NEURAL VS. ETAS, 30-DAY HORIZON, 10
LEAKAGE-FREE WINDOWS. THE GEODETIC-CONTEXT GAIN IS GLOBAL
and REGIONALLY ROBUST.

View	mean IGPE vs. ETAS	windows +	events
Global	+0.115	9/10	2166
Chile	+0.656	9/10	47
Japan	+0.192	8/10	197
California	+1.007	7/9	11
New Zealand	+1.210	8/8	19

the catalog-based test is the correct consistency tool. (This is one 30-day global window; the method generalizes.)

D. The horizon-dependent value of geodetic context

The headline result concerns the strain-conditioned neural challenger. Diagnosing it required first fixing a calibration failure: the raw model over-predicted the 30-day global rate $\approx 490\times$ (98,932 vs. 248 observed) because the training compensator constrained the background only at event locations, leaving the background head free to inflate rate in event-free cells. The fit-time `rate_cal` anchor (computed on the evaluation grid) fixes this: the calibrated model forecasts 218.9 events (ETAS 173.7, observed 248) and passes the N-test. Only the calibrated model is scored below.

At the 1–7 day operational horizon, the geodetic context does not beat ETAS. An 8-window, leakage-free, 7-day back-analysis gives mean IGPE vs. ETAS of -0.053 , positive in only 4/8 windows, with high per-window variance (a single badly-missed sequence window drives -0.80). The ETAS-family stack of Sec. IV-B likewise gains only $+0.009$ globally at 7 days and does not generalize. The 1–7 day mean-rate IGPE is at or near a practical ceiling.

At the 30-day horizon, the same geodetic context beats ETAS robustly and regionally. A 10-window back-analysis (Table IV) gives a global mean IGPE vs. ETAS of $+0.115$ over 2166 events, positive in 9/10 windows, with a paired- t lower bound of $+0.087$ (excludes zero), and it is positive in every high-seismicity region. This is the opposite of the 7-day result.

The mechanism is explicit. The calibrated model’s total forecast is essentially *constant across windows* (18.1 events at 7 days, 77.3 at 30 days, ratio $\approx 30/7$): it deploys a time-flat geodetic *background*, not a triggering model. That is precisely why it loses at 7 days (it cannot track evolving sequences, where ETAS triggering dominates the integral) and wins at

30 days (a better stationary background dominates the longer integral). This is consistent with the global-scale geodetic literature, in which strain rate improves long-term global rate models [14] while its regional short-term contribution is weaker [15]. The honest operational consequence is that the 1–7 day product stays ETAS, and the geodetic context is deployed only as a longer, background-dominated outlook where it measurably earns skill.

V. DISCUSSION

The four results cohere into one message: for global $M \geq 5$ short-term forecasting, base tiled ETAS is at or near the practical ceiling on mean-rate IGPE, and the honest levers lie elsewhere. A separately conducted, adversarial review of 2024–2026 frontier paths (time-varying b , regime-gated neural routing, anisotropic finite-fault kernels, magnitude-head networks, multi-resolution scoring) found that none survived skeptical refutation as a large 1–7 day gain; the convergent verdict matches the EarthquakeNPP conclusion [9] and our own measurements. The value is not in chasing a larger 7-day number but in (i) scoring honestly (catalog-based over-dispersion tests, Sec. IV-C), (ii) deploying a covariate at the horizon where it earns skill (Sec. IV-D), and (iii) refusing to ship a result that does not generalize (Sec. IV-B).

The methodological posture is as much a contribution as any single number. Pre-registered ship rules, shuffled-label negative controls, and the distinction between a well-powered global test (751–2166 events) and an under-powered regional gate (3–54 events) are what separate a real result from a multiple-comparison artifact; the temporally adaptive stack, which a naive read would have shipped, is the clearest example.

VI. HONEST LIMITATIONS

The pseudo-prospective back-analysis spans weekly windows over a limited recent period; the event counts are large (2166 global events at 30 days) but the temporal span is short, so the results are best read as a well-powered snapshot rather than a multi-year prospective record. The neural challenger’s absolute calibration, though fixed by `rate_cal` for scoring, is not production-hardened, and it is not used in the daily 1–7 day product. Only one context channel (GNSS strain) is active; the remaining geophysical covariates are honestly zero-filled and not claimed. The magnitude homogenization uses literature M_w relations as a documented fallback, with a bounded b -value bias that the per-tile fit largely removes. The secondary-cascade under-count is measured on one 30-day global window. None of these is hidden: each corresponds to a flag or an artifact in the released code.

VII. CONCLUSION

We built a leakage-free, multi-region operational earthquake forecasting testbed with pre-registered CSEP evaluation and used it to measure, rather than assert, four things: that ETAS earns skill only where triggering exists; that an ETAS-family stack gains globally but not regionally; that the binding 1–7 day consistency failure is count over-dispersion with a $\approx 28\%$

secondary-cascade under-count; and that the value of a GNSS-strain geodetic covariate is horizon-dependent, absent at 1–7 days and robust at 30 days, because the calibrated model contributes a time-flat background rather than triggering. The practical recommendation is to ship calibrated ETAS at the operational horizon, score with catalog-based over-dispersion tests, and reserve the geodetic context for a longer background-dominated outlook. All code, configurations, provenance manifests, and committed artifacts are released openly so that every number here is reproducible and every negative result is on the record.

DATA AND CODE AVAILABILITY

Code, configurations, provenance manifests, and the committed daily and back-analysis artifacts (including `outlook-evidence-30d.json`, `benchmark.json`, `e12_multiregion.json`, `e13_catalog_based.json`, and the per-region back-analyses) are at https://github.com/fsantibanezleal/CAOS_SEISMIC under the MIT license; a static forecast viewer is live at <https://seismic.fasl-work.com>. Catalog data are from USGS ComCat (public domain) and the Centro Sismológico Nacional, Chile; long-term anchoring from ISC-GEM; mechanisms from the Global CMT project; geodetic velocities from the Nevada Geodetic Laboratory. Their licenses and required attributions are tracked in the repository.

REFERENCES

- [1] R. J. Geller, D. D. Jackson, Y. Y. Kagan, and F. Mulargia, “Earthquakes cannot be predicted,” *Science*, vol. 275, no. 5306, p. 1616, 1997, doi:10.1126/science.275.5306.1616.
- [2] T. H. Jordan *et al.*, “Operational earthquake forecasting: State of knowledge and guidelines for utilization,” *Ann. Geophys.*, vol. 54, no. 4, 2011, doi:10.4401/ag-5350.
- [3] Y. Ogata, “Statistical models for earthquake occurrences and residual analysis for point processes,” *J. Amer. Statist. Assoc.*, vol. 83, no. 401, pp. 9–27, 1988, doi:10.1080/01621459.1988.10478560.
- [4] Y. Ogata, “Space-time point-process models for earthquake occurrences,” *Ann. Inst. Statist. Math.*, vol. 50, no. 2, pp. 379–402, 1998, doi:10.1023/A:1003403601725.
- [5] P. A. Reasenber and L. M. Jones, “Earthquake hazard after a mainshock in California,” *Science*, vol. 243, no. 4895, pp. 1173–1176, 1989, doi:10.1126/science.243.4895.1173.
- [6] M. T. Page, N. van der Elst, J. Hardebeck, K. Felzer, and A. J. Michael, “Three ingredients for improved global aftershock forecasts,” *Bull. Seismol. Soc. Amer.*, vol. 106, no. 5, pp. 2290–2301, 2016, doi:10.1785/0120160073.
- [7] I. Spassiani, G. Falcone, M. Murru, and W. Marzocchi, “Operational earthquake forecasting in Italy: Validation after 10 yr of running,” *Geophys. J. Int.*, vol. 234, no. 3, pp. 2501–2518, 2023, doi:10.1093/gji/ggad256.
- [8] F. Serafini *et al.*, “A benchmark database of ten years of prospective next-day earthquake forecasts,” *Sci. Data*, vol. 12, p. 1501, 2025, doi:10.1038/s41597-025-05766-3.
- [9] S. Stockman, D. J. Lawson, and M. J. Werner, “EarthquakeNPP: Benchmark datasets for earthquake forecasting with neural point processes,” *Trans. Mach. Learn. Res.*, 2026, arXiv:2410.08226.
- [10] P. M. R. DeVries, F. Viégas, M. Wattenberg, and B. J. Meade, “Deep learning of aftershock patterns following large earthquakes,” *Nature*, vol. 560, pp. 632–634, 2018, doi:10.1038/s41586-018-0438-y.
- [11] A. Mignan and M. Broccardo, “One neuron versus deep learning in aftershock prediction,” *Nature*, vol. 574, pp. E1–E3, 2019, doi:10.1038/s41586-019-1582-8.
- [12] K. Dascher-Cousineau, O. Shchur, E. E. Brodsky, and S. Günne-mann, “Using deep learning for flexible and scalable earthquake forecasting,” *Geophys. Res. Lett.*, vol. 50, e2023GL103909, 2023, doi:10.1029/2023GL103909.
- [13] O. Zlydenko *et al.*, “A neural encoder for earthquake rate forecasting,” *Sci. Rep.*, vol. 13, 2023, doi:10.1038/s41598-023-38033-9.
- [14] P. Bird, D. D. Jackson, Y. Y. Kagan, C. Kreemer, and R. S. Stein, “GEAR1: A global earthquake activity rate model constructed from geodetic strain rates and smoothed seismicity,” *Bull. Seismol. Soc. Amer.*, vol. 105, no. 5, pp. 2538–2554, 2015, doi:10.1785/0120150058.
- [15] J. A. Bayona *et al.*, “Prospective evaluation of multiplicative hybrid earthquake forecasting models in California,” *Geophys. J. Int.*, vol. 229, no. 3, pp. 1736–1753, 2022, doi:10.1093/gji/ggac018.
- [16] E. M. Scordilis, “Empirical global relations converting M_S and m_b to moment magnitude,” *J. Seismol.*, vol. 10, pp. 225–236, 2006, doi:10.1007/s10950-006-9012-4.
- [17] S. Wiemer and M. Wyss, “Minimum magnitude of completeness in earthquake catalogs,” *Bull. Seismol. Soc. Amer.*, vol. 90, no. 4, pp. 859–869, 2000, doi:10.1785/0119990114.
- [18] J. Woessner and S. Wiemer, “Assessing the quality of earthquake catalogues: Estimating the magnitude of completeness and its uncertainty,” *Bull. Seismol. Soc. Amer.*, vol. 95, no. 2, pp. 684–698, 2005, doi:10.1785/0120040007.
- [19] J. Zhuang, Y. Ogata, and D. Vere-Jones, “Stochastic declustering of space-time earthquake occurrences,” *J. Amer. Statist. Assoc.*, vol. 97, no. 458, pp. 369–380, 2002, doi:10.1198/016214502760046925.
- [20] I. Zaliapin and Y. Ben-Zion, “Earthquake clusters in southern California I: Identification and stability,” *J. Geophys. Res. Solid Earth*, vol. 118, no. 6, pp. 2847–2864, 2013, doi:10.1002/jgrb.50179.
- [21] G. Blewitt, C. Kreemer, W. C. Hammond, and J. Gazeaux, “MIDAS robust trend estimator for accurate GPS station velocities without step detection,” *J. Geophys. Res. Solid Earth*, vol. 121, no. 3, pp. 2054–2068, 2016, doi:10.1002/2015JB012552.
- [22] P. Bird, “An updated digital model of plate boundaries,” *Geochem. Geophys. Geosyst.*, vol. 4, no. 3, 1027, 2003, doi:10.1029/2001GC000252.
- [23] A. Helmstetter, Y. Y. Kagan, and D. D. Jackson, “High-resolution time-independent grid-based forecast for $M \geq 5$ earthquakes in California,” *Seismol. Res. Lett.*, vol. 78, no. 1, pp. 78–86, 2007, doi:10.1785/gssrl.78.1.78.
- [24] W. H. Savran *et al.*, “pyCSEP: A Python toolkit for earthquake forecast developers,” *Seismol. Res. Lett.*, vol. 93, no. 5, pp. 2858–2870, 2022, doi:10.1785/0220220033.
- [25] W. H. Savran, M. J. Werner, W. Marzocchi, *et al.*, “Pseudoprospective evaluation of UCERF3-ETAS forecasts during the 2019 Ridgecrest sequence,” *Bull. Seismol. Soc. Amer.*, vol. 110, no. 4, pp. 1799–1817, 2020, doi:10.1785/0120200026.
- [26] D. Schorlemmer, M. C. Gerstenberger, S. Wiemer, D. D. Jackson, and D. A. Rhoades, “Earthquake likelihood model testing,” *Seismol. Res. Lett.*, vol. 78, no. 1, pp. 17–29, 2007, doi:10.1785/gssrl.78.1.17.
- [27] J. D. Zechar, M. C. Gerstenberger, and D. A. Rhoades, “Likelihood-based tests for evaluating space-rate-magnitude earthquake forecasts,” *Bull. Seismol. Soc. Amer.*, vol. 100, no. 3, pp. 1184–1195, 2010, doi:10.1785/0120090192.
- [28] D. A. Rhoades *et al.*, “Efficient testing of earthquake forecasting models,” *Acta Geophys.*, vol. 59, no. 4, pp. 728–747, 2011, doi:10.2478/s11600-011-0013-5.
- [29] Y. Y. Kagan, “Earthquake number forecasts testing,” *Geophys. J. Int.*, vol. 211, no. 1, pp. 335–345, 2017, doi:10.1093/gji/ggx300.
- [30] W. Marzocchi, J. D. Zechar, and T. H. Jordan, “Bayesian forecast evaluation and ensemble earthquake forecasting,” *Bull. Seismol. Soc. Amer.*, vol. 102, no. 6, pp. 2574–2584, 2012, doi:10.1785/0120110327.
- [31] M. Herrmann and W. Marzocchi, “Maximizing the forecasting skill of an ensemble model,” *Geophys. J. Int.*, vol. 234, no. 1, pp. 73–87, 2023, doi:10.1093/gji/ggad020.
- [32] T. Gneiting and A. E. Raftery, “Strictly proper scoring rules, prediction, and estimation,” *J. Amer. Statist. Assoc.*, vol. 102, no. 477, pp. 359–378, 2007, doi:10.1198/016214506000001437.
- [33] J. D. Zechar and T. H. Jordan, “Testing alarm-based earthquake predictions,” *Geophys. J. Int.*, vol. 172, no. 2, pp. 715–724, 2008, doi:10.1111/j.1365-246X.2007.03676.x.